Switch-BERT: Learning to Model Multimodal Interactions by Switching Attention and Input

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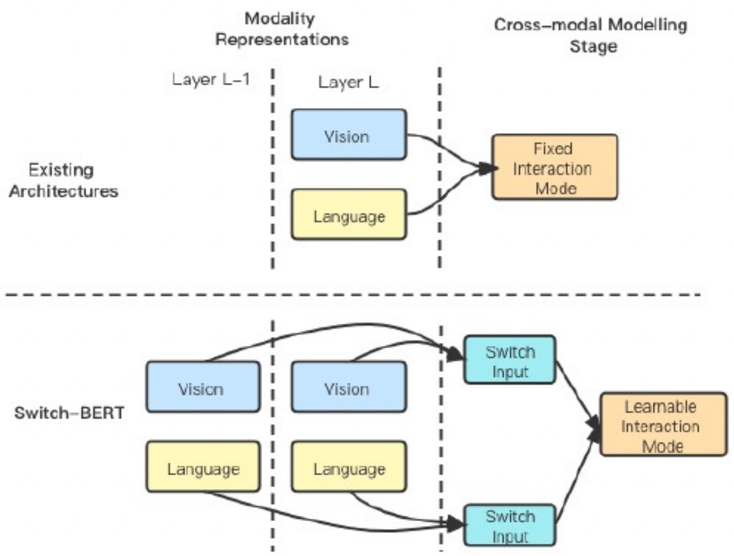
The Modality Mismatch Problem



Existing architectures facing modality mismatch because of **diversity of input modalities** and their **fixed architectures**.

- Diversity of input modalities

single-stream architectures: visual features with high-level semantics and text input with low-level semantics share a common encoding process.

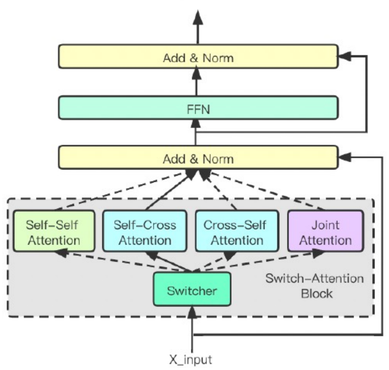
- Fixed architectures

Dual-stream architectures: the interaction between modalities is restricted to specific layers that can be inflexible.

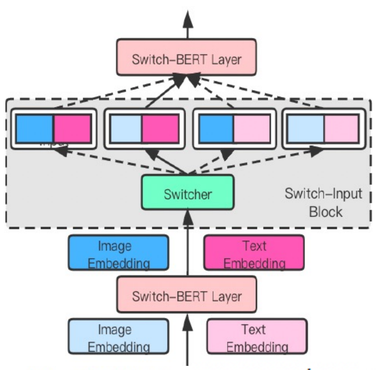
Switch-BERT: mitigates the modality mismatch problem by introducing learnable layer-wise and cross-layer interactions.

Switch-BERT Model

Switch-Attention Block(SAB): learns layer-wise interaction



Switch-Input Block(SIB): introduces cross-layer interaction



Alignment Degree: $z_l^i = AvgPool(X_l^i), z_l^t = AvgPool(X_l^t))).\ \ d_l = z_l^i \odot z_l^t$

Switcher: $\pi = Softmax(f_{MLP}(d_l)).$ $\quad p(M_n) = \frac{\exp((\log(\pi_n) + g_n)/\tau)}{\sum_{j=1}^{N_a} \exp((\log(\pi_j) + g_j)/\tau)}$ $\quad y_{soft} = \sum_{i \in N_a} p(M_i) M_i(X_{input})$

$y_{hard} = M_{n^*}(X_{input}), n^* = \underset{n}{\operatorname{argmax}}\{p(M_n)\}$

Experiments

Using controlled Settings to focus our discussion on the independent contribution of architecture design.

Main Results

Results on three downstream tasks:

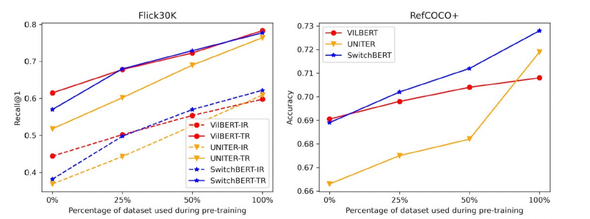
Models		Params	VQAv2	Flick30K-Retrieval Image Retrieval	Flick30K-Retrieval Text Retrieval	RefCOCO+
Single-stream (Fixed)	UNITER[5]	114.9M	68.8±0.4	60.9±0.7	76.4±1.3	71.9±0.67
	VL-BERT[32]	116.1M	68.3±0.31	57.9±1.1	70.9±1.7	71.1±0.23
	VisualBERT[19]	114.9M	68.9±0.27	61.1±1.2	75.5±1.8	69.7±0.31
Dual-stream (Fixed)	LXMERT[34]	211.4M	67.1±0.34	58.6±1.4	74.9±2.7	69.8±0.46
	VilBERT[25]	242.1M	68.7±0.82	59.8±0.8	**78.3**±1.6	70.8±0.58
Dynamic	Switch-BERT*	130.6M	66.7 ±0.97	38.2 ±1.7	57.3 ±2.3	68.9 ±0.82
	Switch-BERT		**70.7**±0.62	**62.2**±0.9	78.2±1.6	**72.8** ±0.45

Ablation Studies

1 Effectiveness of SAB & SIB

Model		Flick30K		VQAv2	RefCOCO+
SIB	SAB	IR(r@1)	TR(r@1)	VQA-score	Accuracy
		60.7	76.2	67.8	69.5
	✓	61.7	76.9	68.9	72.4
✓		60.8	77.7	68.5	71.7
✓	✓	**62.2**	**78.2**	**70.7**	**72.8**

2 Impact on scale of pre-training sets.



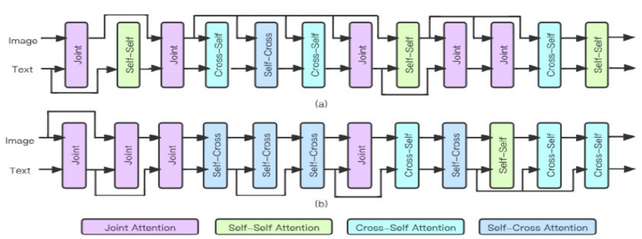
3 Effect of Model's depth

Model	VQAv2	RefCOCO+
#layers	6 → 12	6 → 12
UNITER	64.2 → 68.8	69.7 → 71.9
Switch-BERT	**65.4 → 70.7**	**70.2 → 72.8**
SAB-ONLY	65.0 → 68.9	69.4 → 72.4

4 Computation Overhead and performances.

(Top-K routes are activated)

Models	VQAv2 FLOPs	VQAv2 VQA-score	RefCOCO+ FLOPs	RefCOCO+ Accuracy
UNITER	$2.31*1e^{16}$	68.8	$3.68*1e^{15}$	71.9
VilBERT	$2.72*1e^{16}$	68.7	$4.29*1e^{15}$	70.8
Switch-BERT(K=4)	$8.02*1e^{16}$	70.7	$10.57*1e^{15}$	72.8
Switch-BERT(K=2)	$3.07*1e^{16}$	70.2	$5.27*1e^{15}$	72.1
Switch-BERT(K=1)	$1.97*1e^{16}$	68.2	$3.12*1e^{15}$	70.8

Qualitative Study

Learned architectures:
- sample-specific interactions
- Task-specific architecture